Institute of Geographical Information Systems

CS-250 - Data Structures and Algorithms LAB

Semester: Spring 2026	**Class: SCEE-IGIS - 2024**
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LAB 4: Doubly Linked List

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Task 1: Understanding the Structure

```cpp
#include <iostream>
using namespace std;

struct Node {
    int data;
    Node* next;
    Node* prev;

    Node(int v){
        data=v;
        next=prev=NULL;
    }
};

class DoublyLinkedList{
public:
    Node* head;
    DoublyLinkedList(){ head=NULL; }
};

int main(){
    DoublyLinkedList d;
    cout<<"DLL Initialized";
}
```

```
cd "/Users/alinawaz/Developer/DSA/LAB-04/" && clang++ -std=c++17 task1.cpp -o task1 && ./task1
alinawaz@Alis-MacBook-Air Developer % cd "/Users/alinawaz/Developer/DSA/LAB-04/" && clang++ -std=c++17 task1.cpp -o task1 && ./task1
DLL Initialized%
alinawaz@Alis-MacBook-Air LAB-04 %
```

Task 2: Inserting Carriages

```cpp
#include <iostream>
using namespace std;

struct Node{
 int data;
 Node* next;
 Node* prev;

 Node(int x){
  data=x;
  next=prev=NULL;
 }
};

Node* head=NULL;

void insertFront(int x){
 Node* n=new Node(x);
 if(!head) head=n;
 else{
  n->next=head;
  head->prev=n;
  head=n;
 }
}

void insertEnd(int x){
```

```
alinawaz@Alis-MacBook-Air LAB-04 % cd "/Users/alinawaz/Developer/DSA/LAB-04/" && clang++ -std=c++17 task2.cpp -o task2 && ./task2
30 20 10 40 50 60 %
alinawaz@Alis-MacBook-Air LAB-04 %
```

Task 3: Removing a Carriage

```cpp
#include <iostream>
using namespace std;

struct Node{
 int data; Node* next; Node* prev;
 Node(int x){data=x; next=prev=NULL;}
};

Node* head=NULL;

void insertEnd(int x){
 Node* n=new Node(x);
 if(!head){head=n;return;}
 Node* t=head;
 while(t->next) t=t->next;
 t->next=n; n->prev=t;
}

void deleteFront(){
 Node* t=head;
 head=head->next;
 if(head) head->prev=NULL;
 delete t;
}

void deleteEnd(){
 Node* t=head;
```

```
alinawaz@Alis-MacBook-Air LAB-04 % cd "/Users/alinawaz/Developer/DSA/LAB-04/" && clang++ -std=c++17 task3.cpp -o task3 && ./task3
20 %
alinawaz@Alis-MacBook-Air LAB-04 %
```

Task 4: Searching for a Passenger's Seat

```cpp
#include <iostream>
using namespace std;

struct Node{
 int data; Node* next; Node* prev;
 Node(int x){data=x; next=prev=NULL;}
};

Node* head=NULL;

void insert(int x){
 Node* n=new Node(x);
 if(!head){head=n;return;}
 Node* t=head;
 while(t->next)t=t->next;
 t->next=n; n->prev=t;
}

bool search(int x){
 Node* t=head;
 while(t){
  if(t->data==x) return true;
  t=t->next;
 }
 return false;
}
```

```
alinawaz@Alis-MacBook-Air LAB-04 % cd "/Users/alinawaz/Developer/DSA/LAB-04/" && clang++ -std=c++17 task4.cpp -o task4 && ./task4
1%
alinawaz@Alis-MacBook-Air LAB-04 %
```

Task 5: Reversing the Train Order

```cpp
#include <iostream>
using namespace std;

struct Node{
 int data; Node* next; Node* prev;
 Node(int x){data=x; next=prev=NULL;}
};

Node* head=NULL;

void insert(int x){
 Node* n=new Node(x);
 if(!head){head=n;return;}
 Node* t=head;
 while(t->next)t=t->next;
 t->next=n; n->prev=t;
}

void reverse(){
 Node* c=head;
 Node* t=NULL;
 while(c){
  t=c->prev;
  c->prev=c->next;
  c->next=t;
  c=c->prev;
 }
```

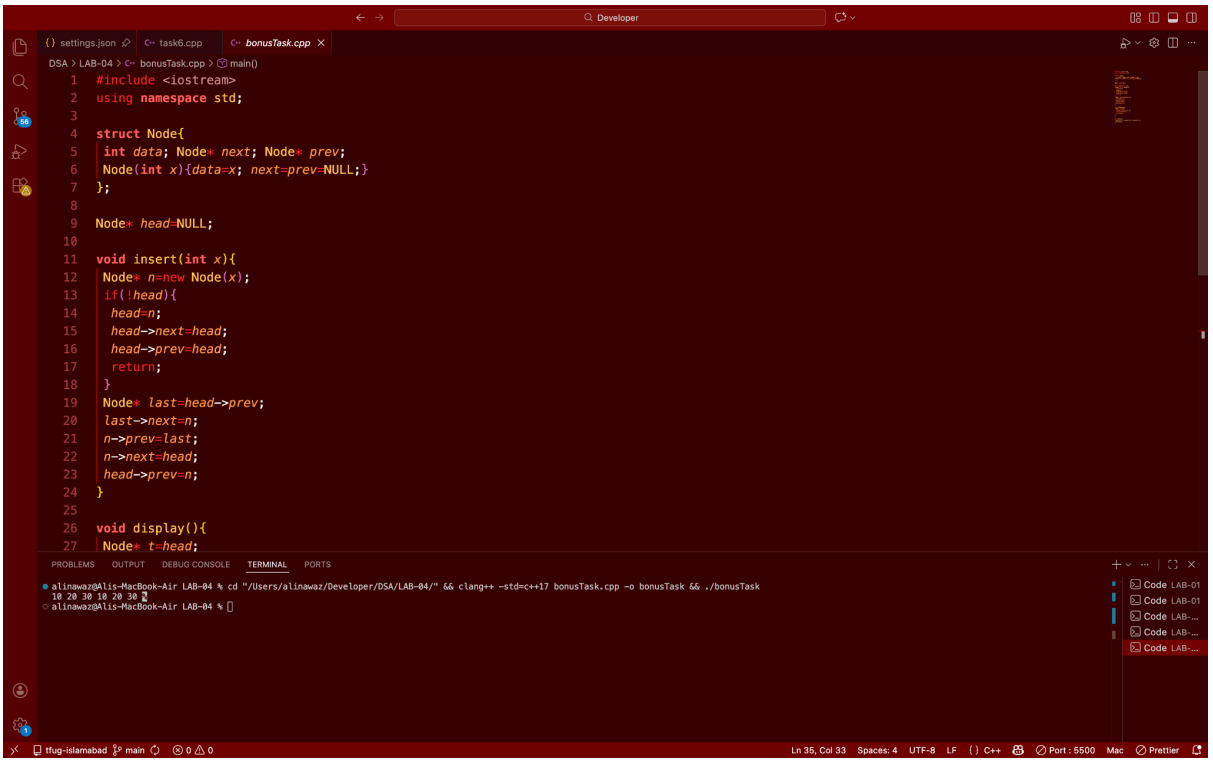
```
alinawaz@Alis-MacBook-Air LAB-04 % cd "/Users/alinawaz/Developer/DSA/LAB-04/" && clang++ -std=c++17 task5.cpp -o task5 && ./task5
30 20 10
alinawaz@Alis-MacBook-Air LAB-04 %
```

Task 6: Booking a Seat Dynamically (User Input)

```cpp
#include <iostream>
using namespace std;

struct Node{
 int data;
 Node* next;
 Node* prev;

 Node(int x){
  data=x;
  next=prev=NULL;
 }
};

Node* head=NULL;

void insert(int x){
 Node* n=new Node(x);
 if(!head){ head=n; return; }

 Node* t=head;
 while(t->next) t=t->next;
 t->next=n;
 n->prev=t;
}

void insertAfter(int k,int v){
```

```
alinawaz@Alis-MacBook-Air LAB-04 % cd "/Users/alinawaz/Developer/DSA/LAB-04/" && clang++ -std=c++17 task6.cpp -o task6 && ./task6
77 10 20 99 30
alinawaz@Alis-MacBook-Air LAB-04 %
```

Bonus Task: Circular Doubly Linked List



```cpp
#include <iostream>
using namespace std;

struct Node{
 int data; Node* next; Node* prev;
 Node(int x){data=x; next=prev=NULL;}
};

Node* head=NULL;

void insert(int x){
 Node* n=new Node(x);
 if(!head){
  head=n;
  head->next=head;
  head->prev=head;
  return;
 }
 Node* last=head->prev;
 last->next=n;
 n->prev=last;
 n->next=head;
 head->prev=n;
}

void display(){
 Node* t=head;
```

```
alinawaz@Alis-MacBook-Air LAB-04 % cd "/Users/alinawaz/Developer/DSA/LAB-04/" && clang++ -std=c++17 bonusTask.cpp -o bonusTask && ./bonusTask
10 20 30 10 20 30 %
alinawaz@Alis-MacBook-Air LAB-04 %
```